

1) Are the conclusions drawn by the reasoner according to your own manually derived conclusions? If not, what could be an explanation? Is the KB consistent? Explain the reasoning.

Initially (before making all individuals different), the reasoner does not make the expected conclusions. One example would be that I expected Mariahilf to be an UrbanNeighborhood. The reasoner already correctly categorized it as a WellLocatedNeighborhood and two Restaurants (LeBurger and Spear) and three Shops (NikeMH, TKMaxxMH and WeekdayMH) were assigned amenities. However, if we don't require the individuals to be different, we could build a model where there is only one restaurant in Mariahilf and both LeBurger and Spear represent that restaurant. This would be a countermodel to the fact that Mariahilf is an UrbanNeighborhood.

Since the HermiT reasoner did not output that the KB is inconsistent, it is consistent.

2) Select 2 axioms from above and write them in DL syntax; also write in DL syntax two inferred axioms (or assertions) and explain the reasoning.

Axioms from above

1. An apartment is located in some residential building or some dual-use building: $\text{Apartment} \sqsubseteq \exists \text{located-in.}(\text{DualUse} \sqcup \text{ResidentialBuilding})$
2. A person lives in some apartment.: $\text{Person} \sqsubseteq \exists \text{lives-in.Apartment}$

Inferred Axioms/Assertions

1. Given the axiom " $\text{Building} \equiv \text{CommercialBuilding} \sqcup \text{DualUse} \sqcup \text{Industrial} \sqcup \text{ResidentialBuilding}$ ", the reasoner can infer " $\text{DualUse} \sqsubseteq \text{Building}$ ". In other words, if something is a Building iff it is a CommercialBuilding, a DualUse, Industrial or ResidentialBuilding, then we can infer that a DualUse is a building.
2. The assertion "Owner(Bob)" is inferred. Since Bob is a Person ($\text{Person}(\text{Bob}) \in \text{ABox}$) and owns apartment Apt2 ($\text{owns}(\text{Bob}, \text{Apt2}) \in \text{ABox}$).

3) Select Edit/Make all individuals different... and start the reasoner again. What happens? Is your KB still consistent? What are the results now? Are the conclusions that are drawn by the reasoner now equal to your own conclusions? Explain the reasoning.

Yes, the KB is still consistent, since the HermiT reasoner did not output that the KB is inconsistent. Now we get the initially expected results (e.g. that Mariahilf is an UrbanNeighborhood), since I first implicitly assumed all individuals to be distinct (e.g. LeBurger $\neq$ Spear).

4) How does the inferred hierarchy differ from the one you explicitly set up? Are there any residential neighborhoods? If not, what could be the explanation?

I did not set up any hierarchy explicitly. Protege was able to syntactically infer that

- DesirableApartment and LeasedApartment are subclasses of Apartment,
- UrbanNeighborhood is a subclass of WellLocatedNeighborhood,
- FamilyNeighborhood, ResidentialNeighborhood and WellLocatedNeighborhood are subclasses of Neighborhood,

- Landlord is a subclass of Owner and that
- Owner and Tenant are subclasses of Person.

However, to show that

- School, Shop, Park, Restaurant and MetroStation are subclasses of Amenity and that
- ResidentialBuilding, CommercialBuilding, DualUse and Industrial are subclasses of Building Protege needed the reasoner.

There are no residential neighborhoods because of the condition (not (contains some Industrial)). It does not follow from the KB that any neighborhood cannot contain industrial buildings. We could create a model of the KB where each neighborhood contains an industrial building. Hence, it does not follow from the KB that any of the given neighborhoods is residential.

AI Disclosure

I did use AI to understand some features of Protege. For instance, I was confused about the following situation in the Classes tab: Protege displayed Owner as a subclass of Person after I added the "Person and (owns some Apartment)" constraint, but it did not display the specific building types as subclasses of building after adding the constraint "CommercialBuilding or DualUse or Industrial or ResidentialBuilding" to Building. AI helped me understand that Protege does not use the reasoner results but only simple syntactic checks for inferring subclasses in the Classes tab.

AI also helped me understand the terminology used in Protege (i.e. that Class = Concept, etc.).